

Executive Summary: Heritage Masonry Skill Assessment & Inspection App

A comprehensive architectural and functional report on the end-to-end apprentice diagnostic and instructor grading workstation built today.

EXECUTIVE MISSION & IMPACT

Today, the platform was transformed from a static masonry database into an **active clinical apprenticeship diagnostic platform**. Instructors can ingest authentic masonry specimens from the field via camera QR, calibrate ground-truth tolerance hit zones, and leverage a dual-mode AI defect engine. Apprentices log in with zero friction via email and a 4-digit PIN, mark defects interactively, advance through curated specimens, and benchmark their accuracy against real-time class heatmaps.

Core System Pillars Built Today

• Pillar 1: Photographic Guidelines & Mobile QR Capture

- **Inspection Photography HUD:** Enforces strict standards across 4 parameters: 90° Orthogonal Plane, Diffuse Ambient Lighting, Macro-to-Course Scope, and 1080p+ Resolution.
- **Instant Mobile QR Pairing:** Instructors scan a dynamic desktop QR code to open the phone camera without entering credentials, using cryptographically signed timed tokens (URLSafeTimedSerializer).
- **Dual Cloud & Local Storage:** Saves full-resolution originals locally and streams to Cloudinary CDN for instant multi-device rendering.

• Pillar 2: Dual-Mode AI Defect Suggestion Engine & 'AI Reviewed' Tagging

- **Immediate Suggestion on Ingestion:** Image uploads automatically trigger AI defect detection, pre-populating normalized pins (x, y) with clinical titles, severity, tolerances, and conservation remedies.
- **Dual-Engine Reliability:** Direct REST integration with Google Gemini 1.5 Flash when API key is set; intelligent clinical masonry rules fallback across all 7 archetypes (lime mortar, dry stone, ashlar, rubble, etc.) when offline or unkeyed.
- **'AI Reviewed' Verification Badge:** System tags analyzed specimens with is_ai_reviewed = True, displaying a high-visibility badge in the admin queue and grader workstation header.
- **Full Grader Authority:** Instructors retain complete control to adjust coordinates, change defect modes, delete false positives, or add custom targets before saving authoritative ground truth.

• Pillar 3: Ground-Truth Defect Grader Workstation

- **Interactive Hit-Zone Calibration:** Graders place numbered pins and calibrate tolerance radii (4% to 15% of wall dimension) with live dashed tolerance circle previews on canvas.
- **Specimen-Specific Defect Dropdown Palette:** Eliminates student confusion by prioritizing relevant modes at the top of the menu while supporting custom in-situ defect title creation.
- **Conservation Action Prescriptions:** Ties every ground-truth defect to standard remedies (e.g. repoint_lime, helical_stitch, structural_pinning, stone_indent).

• Pillar 4: Apprentice Point-and-Tag Workstation & Sequential Advancement

- **Interactive Orthogonal Canvas:** High-resolution pan and zoom controls allowing candidates to drop defect pins directly on suspect arrises, bed joints, and rubble voids.
- **Multi-Tier Euclidean Scoring Engine:** Evaluates student pins against ground truth using distance tolerances: Full Hits (100% points), Partial Mode Hits (50% points), False Positives, and Missed Defect penalties.
- **Sequential Advancement:** 'Next Specimen' progression flow automatically passes candidate identity and progress across the curriculum sequence (Specimen 1 → Specimen 2 → Specimen 3 → Portfolio).
- **Dashboard Completion Indicators:** Specimen cards reflect real-time completion status and candidate score percentage.

• Pillar 5: Student Email Enrollment & 4-Digit PIN Authentication

- **Low-Friction Enrollment:** Candidates enter email, name, and choose a memorable 4-digit PIN (defaults to 0000), eliminating complex password reset friction on job sites.
 - **Student Portal Dashboard (/portal):** Unified candidate dashboard listing all completed specimen assessments, dates, score gauges, pass/fail status, and direct review links.
- **Pillar 6: Thermal Cohort Heatmap Overlay & Class Analytics**
 - **Multi-Stop Thermal Gradient Canvas:** Client-side HTML5 canvas overlay rendering radial density glows (Cyan → Yellow → Orange → Hotspot Red) using lighter composite blending.
 - **Target Hit-Rate Benchmarking:** Computes real-time percentage of the class that successfully spotted each target (e.g., '88.9% hit rate on Bed Joint Washout; 50% on Spalling').
 - **Dual-Perspective Utility:** Accessible to students post-evaluation for peer benchmarking, and to instructors to identify misleading masonry features or common student misconceptions.

Verification & Automated Test Suite Results

The platform was verified through automated test suites covering all full-stack user journeys, database schemas, and mathematical hit-testing algorithms:

Test Suite Module	Core Features Verified	Tests	Result
test_advancement_and_scoring.py	Sequential progression, next specimen links, email/PIN auth, student portal scoring	9	PASSED
test_ai_suggestions.py	Gemini/fallback AI suggestions, upload auto-trigger, 'AI Reviewed' tagging, grader edit/save	4	PASSED
test_cohort_heatmap.py	Spatial density aggregation, hit-rate percentage math, cohort filtering, HTML5 canvas markup	4	PASSED
Total Full-Stack Verification	Comprehensive regression, database migration, and user flow coverage	17	17 / 17 OK

Production Deployment & Git Milestones

Commit	Scope of Changes	Branch
152c975	Student email enrollment, 4-digit PIN authentication, sequential progression, and student portal	origin/main
8a044a7	Dual-engine AI defect suggestions on upload and 'AI Reviewed' tagging in grader & admin queue	origin/main
93a759e	Cohort inspection heatmap overlay for student and grader workstations with target hit rates	origin/main

Next Horizons & Strategic Recommendations

- **Tiered Apprentice Hint System:** Contextual structural clues from the 'Master Mason' with minor score deductions.
- **Interactive Conservation 'After-State':** Visual slider demonstrating correct lime pointing vs. destructive cement ribbons.
- **BS 7913 / ICOMOS Survey PDF Export:** 1-click generation of professional 1-page condition survey reports.
- **Guild Mastery Digital Passport:** Verifiable competency credentials upon achieving 80%+ mastery across all 7 archetypes.